6.20 (a) In Coulomb gauge,

$$\Phi(\vec{x},t) = \frac{1}{4\pi\epsilon_0}\int d^3x' \frac{\rho(\vec{x}',t)}{|\vec{x}-\vec{x}'|} = \frac{1}{4\pi\epsilon_0}\int d^3x' \frac{1}{|\vec{x}-\vec{x}'|}\delta(x')\delta(y')\delta'(z')\delta(t)$$

$$= \frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty} dz' \frac{\delta'(z')}{[x^2+y^2+(z-z')^2]^{1/2}} = -\frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty} dz'\,\delta(z')\frac{\partial}{\partial z'}\left[x^2+y^2+(z-z')^2\right]^{-1/2}$$

$$= -\frac{\delta(t)}{4\pi\epsilon_0}\int_{-\infty}^{+\infty} dz'\,\delta(z')\frac{z-z'}{[x^2+y^2+(z-z')^2]^{3/2}} = -\frac{1}{4\pi\epsilon_0}\delta(t)\frac{z}{r^3}$$

(b) The transverse current is defined as $\vec{J}_t(\vec{x},t) = \frac{1}{4\pi}\nabla\times\nabla\times\int d^3x' \frac{\vec{J}(\vec{x}',t)}{|\vec{x}-\vec{x}'|}$

The integration can be performed as $\int d^3x' \frac{\vec{J}(\vec{x}',t)}{|\vec{x}-\vec{x}'|} = -\delta'(t)\frac{\vec{\epsilon}_3}{r}$. Then

$$\vec{J}_t(\vec{x},t) = \frac{1}{4\pi}\nabla\times\nabla\times\vec{J}(\vec{x},t) = -\frac{1}{4\pi}\delta'(t)\left[\nabla\left(\nabla\cdot\frac{\vec{\epsilon}_3}{r}\right) - \nabla^2\left(\frac{\vec{\epsilon}_3}{r}\right)\right].$$

The first term can be written as

$$\nabla\left(\nabla\cdot\frac{\vec{\epsilon}_3}{r}\right) = \nabla\left(\vec{\epsilon}_3\cdot\nabla\left(\frac{1}{r}\right)\right) = \nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right),$$

and the second term is $\nabla^2\left(\frac{\vec{\epsilon}_3}{r}\right) = \vec{\epsilon}_3\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta(\vec{x})\vec{\epsilon}_3$. Then,

$$\vec{J}_t(\vec{x},t) = -\frac{1}{4\pi}\delta'(t)\left[\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) + 4\pi\delta(\vec{x})\vec{\epsilon}_3\right] = -\delta'(t)\left[\vec{\epsilon}_3\delta(\vec{x}) + \frac{1}{4\pi}\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)\right]$$

Notice that $\frac{\partial}{\partial z}\left(\frac{1}{r}\right) = -\frac{z}{r^3}$, and

$$\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) = -\nabla\left(\frac{z}{r^3}\right) = -\vec{\epsilon}_i\partial_i\left(\frac{z}{r^3}\right) = \frac{\vec{\epsilon}_i\cdot 3x_i z}{r^5} - \frac{\vec{\epsilon}_3}{r^3} = \frac{3\vec{x}(\vec{\epsilon}_3\cdot\vec{x})}{r^5} - \frac{\vec{\epsilon}_3}{r^3} = \frac{3\vec{n}(\vec{\epsilon}_3\cdot\vec{n}) - \vec{\epsilon}_3}{r^3},$$

for $r\neq 0$. Therefore, $J_t(\vec{x},t) = -\delta'(t)\left[\vec{\epsilon}_3\delta(\vec{x}) + \frac{3\vec{n}(\vec{\epsilon}_3\cdot\vec{n}) - \vec{\epsilon}_3}{4\pi r^3}\right]$, $r\neq 0$.

To account for the singularity at $r\neq 0$, consider the integral of the second term of the transverse current over a ball with radius R and center at origin, $\vec{x}=0$, then

$$\frac{1}{4\pi}\int d^3x\,\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) = \frac{1}{4\pi}\oint_S \frac{\partial}{\partial z}\left(\frac{1}{r}\right)\vec{n}\,da = -\frac{1}{4\pi}\oint\frac{\vec{\epsilon}_3\cdot\vec{n}}{r^2}\vec{n}\,da = -\frac{1}{4\pi}\int(\vec{\epsilon}_3\cdot\vec{n})\vec{n}\,d\Omega$$

This can be shown to be $-\frac{1}{3}\vec{\epsilon}_3$ and we can write $\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right)$ as $\nabla\left(\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right) = -\frac{4\pi}{3}\vec{\epsilon}_3\delta(\vec{x}) + \frac{3(\vec{\epsilon}_3\cdot\vec{n})\vec{n}-\vec{\epsilon}_3}{r^3}$.

Therefore, $\vec{J}_t(\vec{x},t) = -\delta'(t)\left[\frac{2}{3}\vec{\epsilon}_3\delta(\vec{x}) + \frac{3(\vec{\epsilon}_3\cdot\vec{n})\vec{n}-\vec{\epsilon}_3}{4\pi r^3}\right]$.

(c) From part (a), $\frac{\partial \Phi}{\partial t} = -\frac{1}{4\pi\varepsilon_0}\delta'(t)\frac{z}{r^3}$. Using the result from part (b), we have

$$\frac{1}{c^2}\nabla\left(\frac{\partial\Phi}{\partial t}\right) = -\frac{\mu_0}{4\pi}\delta'(t)\nabla\left(\frac{z}{r^3}\right) = \frac{\mu_0}{4\pi}\delta'(t)\nabla\left[\frac{\partial}{\partial z}\left(\frac{1}{r}\right)\right] = \frac{\mu_0}{4\pi}\delta'(t)\left[-\frac{4\pi}{3}\vec{\varepsilon}_3\delta(\vec{x}) + \frac{3\vec{\varepsilon}_3(\vec{\varepsilon}_3\cdot\vec{n}) - \vec{\varepsilon}_3}{r^3}\right]$$

$$= -\mu_0\delta'(t)\left[\frac{1}{3}\vec{\varepsilon}_3\delta(\vec{x}) + \frac{\vec{\varepsilon}_3}{4\pi r^3} - \frac{3}{4\pi r^3}\vec{n}(\vec{\varepsilon}_3\cdot\vec{n})\right]$$

On the other hand,

$$\vec{J}_l = \vec{J} - \vec{J}_t = -\delta'(t)\left[\frac{1}{3}\vec{\varepsilon}_3\delta(\vec{x}) + \frac{\vec{\varepsilon}_3}{4\pi r^3} - \frac{3}{4\pi r^3}\vec{n}(\vec{\varepsilon}_3\cdot\vec{n})\right]$$

Therefore, $\frac{1}{c^2}\nabla\left(\frac{\partial\Phi}{\partial t}\right) = \mu_0\vec{J}_l$, and we only need $\vec{J}_t$'s contribution to the EM fields. In the Coulomb gauge the vector potential is given by

$$\vec{A}(\vec{x},t) = \frac{\mu_0}{4\pi}\int d^3x'\,\frac{1}{R}\left[\vec{J}_t(\vec{x}',t')\right]_{ret} = -\frac{\mu_0}{4\pi}\int d^3x'\,\frac{1}{R}\delta'(t-R/c)\left[\vec{\varepsilon}_3\delta(\vec{x}') + \frac{1}{4\pi}\nabla'\left\{\frac{\partial}{\partial z'}\left(\frac{1}{r'}\right)\right\}\right]$$

The first integral can be easily done, but I have no idea how to do the second. So, I will calculate things in Lorenz gauge to show the electric field components.

Using Eq (6.47), the scalar potential is

$$\Phi(\vec{x},t) = \frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\vec{x}',t-|\vec{x}-\vec{x}'|/c)}{|\vec{x}-\vec{x}'|}d^3x' = \frac{1}{4\pi\varepsilon_0}\int\frac{\delta'(z')}{[x^2+y^2+(z-z')^2]^{1/2}}\delta\left(t-\sqrt{x^2+y^2+(z-z')^2}/c\right)dz'$$

$$= -\frac{1}{4\pi\varepsilon_0}\int\delta(z')\frac{\partial}{\partial z'}\left[\frac{1}{[x^2+y^2+(z-z')^2]^{1/2}}\delta\left(t-\sqrt{x^2+y^2+(z-z')^2}/c\right)\right]dz'$$

$$= -\frac{1}{4\pi\varepsilon_0}\int\delta(z')\left\{\frac{z-z'}{[x^2+y^2+(z-z')^2]^{3/2}}\delta\left(t-\sqrt{x^2+y^2+(z-z')^2}/c\right)\right.$$
$$\left. + \frac{1}{[x^2+y^2+(z-z')^2]^{1/2}}\delta'\left(t-\sqrt{x^2+y^2+(z-z')^2}/c\right)\frac{z-z'}{c[x^2+y^2+(z-z')^2]^{1/2}}\right\}dz'$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{z}{r^2}\left(\frac{1}{r}\delta(t-r/c) + \frac{1}{c}\delta'(t-r/c)\right)$$

and vector potential is

$$\vec{A}(\vec{x},t) = \frac{\mu_0}{4\pi}\int\frac{\vec{J}(\vec{x}',t-|\vec{x}-\vec{x}'|/c)}{|\vec{x}-\vec{x}'|}d^3x' = -\frac{\mu_0}{4\pi}\frac{\delta'(t-r/c)}{r}\vec{\varepsilon}_3$$

It is straightforward to verify the Lorenz condition is valid.

The electric field can be directly calculated as

$$\vec{E} = -\nabla\Phi - \frac{\partial \vec{A}}{\partial t} = \frac{1}{4\pi\varepsilon_0}\nabla\left[\frac{z}{r^3}\delta(t-r/c) + \frac{z}{cr^2}\delta'(t-r/c)\right] + \frac{\mu_0}{4\pi}\frac{\delta''(t-r/c)}{r}\vec{\varepsilon}_3$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\nabla\left(\frac{z}{r^3}\delta(t-r/c) + \frac{z}{cr^2}\delta'(t-r/c)\right) + \frac{\delta''(t-r/c)}{c^2 r}\vec{\varepsilon}_3\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{-3\vec{n}(\vec{n}\cdot\vec{\varepsilon}_3)+\vec{\varepsilon}_3}{r^3}\delta(t-r/c) - \frac{z\vec{r}}{cr^4}\delta'(t-r/c)\right.$$
$$\left. + \frac{-2\vec{n}(\vec{n}\cdot\vec{\varepsilon}_3)+\vec{\varepsilon}_3}{cr^2}\delta'(t-r/c) - \frac{z\vec{r}}{c^2r^3}\delta''(t-r/c) + \frac{\delta''(t-r/c)}{c^2 r}\vec{\varepsilon}_3\right]$$

Notice that $\delta(t-r/c) = c\delta(r-ct)$, $\delta'(t-r/c) = -c^2\delta'(r-ct)$, $\delta''(t-r/c) = c^3\delta''(r-ct)$, then we can write down the components as

$$E_x = \frac{1}{4\pi\varepsilon_0}\left[-\frac{3\sin\theta\cos\phi\cdot\cos\theta}{r^3}c\delta(r-ct) + \frac{\cos\theta\sin\theta\cos\phi}{cr^2}c^2\delta'(r-ct)\right.$$
$$\left. + \frac{2\sin\theta\cos\phi\cos\theta}{cr^2}c^2\delta'(r-ct) - \frac{\cos\theta\sin\theta\cos\phi}{c^2 r}c^3\delta''(r-ct)\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{c}{r}\left[-\delta''(r-ct) + \frac{3}{r}\delta'(r-ct) - \frac{3}{r^2}\delta(r-ct)\right]\sin\theta\cos\theta\cos\phi,$$

and E_y can be obtained from E_x by replacing $\cos\phi$ with $\sin\phi$. With

$$E_z = \frac{1}{4\pi\varepsilon_0}\left[\frac{-3\cos^2\theta+1}{r^3}c\delta(r-ct) + \frac{\cos^2\theta}{cr^2}c^2\delta'(r-ct)\right.$$
$$\left. - \frac{-2\cos^2\theta+1}{cr^2}c^2\delta'(r-ct) - \frac{\cos^2\theta}{c^2 r}c^3\delta''(r-ct) + \frac{1}{c^2 r}c^3\delta''(r-ct)\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{c}{r}\left[\sin^2\theta\,\delta''(r-ct) + (3\cos^2\theta-1)\left(\frac{1}{r}\delta'(r-ct) - \frac{1}{r^2}\delta(r-ct)\right)\right]$$

Clearly these components are caused.